THREE INCLUSIONS DESCRIBING THE COMMUTATOR OF THE ALGEBRA O IN TERMS OF THE IDEAL GENERATED BY AB MINUS BA

PAUL TERWILLIGER

Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/sigma-2011-069-askey-wilson/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the retained preamble is the source's own layout adapter plus the author's macro block at lines 10-18 of sources/sigma-2011-069-askey-wilson/pinned-original-source.tex (everything between the publisher class metadata and \begin{document}, comments excluded). Nothing was added, no macro was invented, and the publisher document class is deliberately not reproduced; the excerpt serves only as preamble, never as a span coordinate.

Editorial note. External references are navigation pointers only; every deliberately kept step of the author's argument stays inside the counted spans. The displayed numbers of those pointers are the author's own numbering under the author's \newtheorem declarations (the source has no subsections, so numbering is section.index), computed mechanically by the lane's scanner.

Editorial note. The single reference to another labelled item is Theorem prop:altpres, recorded as a navigation stub of kind theorem carrying the author's own number; the computation that consumes it stays inside the counted span.

Editorial note. Clause (iii) is closed by the author in one line from clauses (i) and (ii) together with centrality of gamma; that sentence is kept verbatim.

Editorial note. Licence binding: the card's source_version now carries the value recorded in this source's own licence evidence (sources/sigma-2011-069-askey-wilson/license-observation.json), namely 'SIGMA7(2011),069; publisher final form2011-07-09'. The adopt step of the publication line compares the two and refused the earlier revision; the card is aligned to the evidence rather than the evidence to the card.

Allowed standard prerequisites (editorial).

- The algebra O is generated by A and B. Same source, set up before the statement; used inside the counted proof at lines 3070-3073. Makes the quotient O/R commutative, which is the whole content of clause (i).
- The alternating presentation relation of the algebra, third displayed equation. Same source, Theorem labelled prop:altpres; invoked inside the counted proof at lines 3081-3085. Identifies the right-hand side of that relation with a nonzero scalar multiple of [A,B] gamma, which gives clause (ii).

Source reference. Original theorem 1.6: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 3082–3082. Navigation pointer for \ref{prop:altpres} used inside the counted span (line 3082); the argument that consumes it is retained in the span.

Lemma 0.1. *The following* (*i*)–(*iii*) *hold.*

Date: doi:10.3842/SIGMA.2011.069; CC BY4.0.

(*i*) $[\mathbb{O}, \mathbb{O}] \subseteq \mathbb{O}[A, B]\mathbb{O}$.
(*ii*) $[A, B]\gamma \in [\mathbb{O}, \mathbb{O}]$.
(*iii*) $\mathbb{O}[A, B]\mathbb{O}\gamma \subseteq \mathbb{O}[A, B]\mathbb{O}$.

Proof. (*i*) Abbreviate $R = \mathbb{O}[A, B]\mathbb{O}$ and consider the quotient algebra $\mathbb{O}/R$. The elements A, B generate $\mathbb{O}$, and these generators satisfy $[A, B] \in R$. Therefore $A + R$, $B + R$ generate $\mathbb{O}/R$, and these generators commute. This shows that $\mathbb{O}/R$ is commutative. Consider the canonical map $\mathbb{O} \to \mathbb{O}/R$. This map has kernel R. The map sends $[\mathbb{O}, \mathbb{O}] \mapsto 0$ since $\mathbb{O}/R$ is commutative. Therefore $[\mathbb{O}, \mathbb{O}] \subseteq R$.

(*ii*) In the third displayed equation of Theorem 1.6, the expression on the right is a nonzero scalar multiple of $[A, B]\gamma$. The expression on the left is equal to $[A^2, B^2] + (q^2 + q^{-2})[B, ABA]$ and is therefore in $[\mathbb{O}, \mathbb{O}]$. The result follows.

(*iii*) By (*i*), (*ii*) above and since γ is central. □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.